

Kayoon Kim

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RESEARCH VISION

My research vision is to design AI and algorithms that serve humanity by facilitating interactions between people and AI systems in everyday life. My research has been focused on (1) exposing social problems and developing data-driven solutions, (2) enhancing human decision-making while preventing marginalization.

Research Interest: Human-AI Interaction, Social Computing, Computational Social Science

EDUCATION

University of Tübingen, Tübingen, Germany

2023.10-2026.03

M.S., Quantitative Data Science, *Magna cum laude*

Advisors: Jan Henry Belz, Prof. Augustin Kelava

Examiners: Dr. Tobias Grosse-Puppenthal, Prof. Holger Brandt

Thesis: Carmony: Multimodal Context-Aware Device Orchestration in Multi-User Automotive Environments (Tentative)

Chung-Ang University, Seoul, South Korea

2017.03-2023.02

B.A. B.E., Sociology (Major), Computer Science (Concentration), *Cum laude*

Advisor: Prof. Chan S. Suh

Thesis: Claiming for Rights: The Changing Landscape of Issues and Claims in the National Petition of South Korea, 2017-2020

RESEARCH EXPERIENCE

Master Thesis Student, Porsche AG Human-Centered AI Research Lab

2025

• Advisor: Jan Henry Belz

Implementing a context-aware system using multi-agent architectures for decision-making on optimal vehicle configurations and seamless human-vehicle interaction for all occupants

Research Intern, Bosch Center for Artificial Intelligence (BCAI), Germany

2024

• Advisor: Lukas Grossberger

Developed and tested Preferential Bayesian Optimization algorithms to efficiently optimize machinery settings for expert operators, modeling human preferences in decision-making processes and identifying cognitive biases for future human-in-the-loop experiments

Research Assistant, Methods Center, University of Tübingen

2023-2024

• Advisor: Prof. Augustin Kelava

Developed the model to predict student dropouts in STEM classes using multivariate time-series analysis (i.e., Kalman Filter) to strengthen the educational environment

Research Assistant, [HIKE Lab](#): Team Cayley, Chung-Ang University

2020-2021

• Advisor: Prof. Haklae Kim

[COVID-19: Our Memory](#): Investigated pandemic-driven social inequalities through statistical correlation analysis, text mining, and news media archiving; awarded at the hackathon sponsored by the Ministry of Interior and Safety of Korea for the project

COVID-19 Data Analysis A to Z: Assessed policy efficacy (i.e., screening facilities, hospital bed capacity), enhanced data visualization accessibility for the general public, and developed recommendations for public health data infrastructure improvements

As a first case of a non-governmental digital archive, projects were donated to the National Library of Korea

	Research Assistant, Da Move Lab, Chung-Ang University	2020-2023
	• Advisor: Prof. Chan S. Suh Collected and analyzed 3-year dataset of South Korea's online petitions using text mining and NLP techniques, revealing embedded biases including political polarization, patriarchal attitudes, and cultural lags that could corrupt AI training data; presented findings at International Postgraduate and Academic Conference 2021 Extended analysis to 5-year dataset, employing fine-tuned language models for sentiment analysis and regression modeling to identify structural limitations in online petition systems, demonstrating how the system and certain cultural norms restrict civic engagement	
PAPERS	[2] Carmony: Multimodal Context-Aware Device Orchestration in Multi-User Automotive Environments (Tentative) Jan Henry Belz*, Kayoon Kim* <i>In progress, UIST 2026</i> *Equal Contribution [1] Mobilizing grievances in the internet age: The case of national online petitioning in South Korea, 2017–2022 Kayoon Kim, Chan S. Suh PLoS ONE 2024	
WORK EXPERIENCE	Data Quality Manager Intern, Kakao Style, South Korea	2023
	• Analyzed model prediction data and developed dashboards to enhance model accuracy and user experience. Designed and implemented an automatic labeling model to streamline the workflow	
HONORS & GRANTS	DAAD STIBET Scholarship , German Academic Exchange Service (DAAD)	2025
	Research Grant , Porsche AG	2025
	Research Grant , National Research Foundation of Korea	2024
	Merit-and-need-based Scholarship , Seoul Scholarship Foundation	2022
	Merit-and-need-based Scholarship , Lotte Scholarship Foundation	2021
	Community Award , Ministry of the Interior and Safety, South Korea	2020
	National Scholarship (10 semesters) , Korea Student Aid Foundation	
SERVICE	Communication Chair Volunteer, Tübingen Women in Machine Learning	2025
	Content Creator Volunteer, Soapbox Science Tübingen	2025
	Guest Speaker, Chung-Ang University	2022
	• Delivered a single session lecture on the overview of computational social science and data science for an Introduction to Sociology course to engage freshmen	
	Academic Advisor, COSADAMA	2020-2021
	• Designed instructional sessions on Git, GitHub, and Python programming for non-STEM major students; mentored participants throughout the learning process	
SKILLS	Language English (Full professional proficiency), Korean (Native) Programming and Development Python, R, C, SPSS, MySQL, Pytorch, Git, Linux	